Chaos Pendulum Site Fault Check

Symptom: controller runs, relays click, arm moves only weakly. Work top to bottom. Outside the wall first, open it last.

Bring

- Multimeter + probes
- Small flat and Phillips screwdrivers, torch
- Marker pen (for slip test)
- Phone with charge: **photograph everything** before changing it

Safety. The VFD holds lethal voltage for **10 min after mains off** (VersiDrive manual). Isolate, wait, then measure 0 V before touching U/V/W or L/N. When the brake releases, the arm can swing. Keep clear.
Do not change any VFD parameter before photographing it.

Shortcut: where to look

Kick shows…	Go to
Fault code on display	A
Hz rises slowly, current low	C (parameters)
Current very high	D (brake), then F
Hz and current normal, arm weak	F (mechanical slip)
Motor hums or shakes	E (wiring)

A · VFD display

Power on, controller running, touch nothing yet.

- [ ] Photo of the display. It shows `StoP`, `H xx.x` (Hz), or a fault code.
  Display: _____
- [ ] **Fault log P-13:** at `StoP`, hold **Enter >1 s**. The display shows `P-01`. Use ▲ to reach `P-13`, then hold **Enter >1 s**. ▲▼ step through the last 4 trips (newest first). Hold **Enter >1 s** twice to exit.
  Trips: _____ _____ _____ _____

Common trip codes

These are the usual codes for this drive family (Invertek Optidrive E2, which the VersiDrive i E2 is based on). Your manual has no code list, so treat them as a guide.

Code	Meaning → likely cause here
`O-I`, `h O-I`	Overcurrent → brake dragging, jam, short in motor cable
`I.t-trP`	Motor overload over time → brake dragging, or P-08 set too low
`O-Volt`	DC bus overvoltage → braking too hard (P-04 short), arm driving the motor
`U-Volt`	Mains low or switched off. Normal after a power cut.
`O-t`	Drive too hot → ventilation, overload
`P-dEF`	Factory defaults loaded → **settings were lost**, go to C
`E-trip`	External trip input / thermistor

B · Watch a kick

Let the controller run. A kick is K2 (MOTOR IMPULSE) clicking.

- [ ] While the drive runs, the display shows **Hz**. Peak Hz during a kick: _____
- [ ] Short press **Enter** (navigate) to show **A** (motor current). Peak A during a kick: _____
- [ ] Motor rated current (nameplate, or P-08): _____ A
- [ ] Listen at the wall during the kick: **hum** / **growl** / **click** / **silence**

Result	Points to
Hz barely rises before the kick ends, current low	**C**: ramp too long or speed set low
Current ≥ 1.5× rated, Hz struggles	**D**: brake not releasing; else **F**
Hz and current look normal	**F**: motor turns, arm slips
Hum, shaking, uneven current	**E**: phase missing

C · Parameters

Enter the menu as in A. **Photograph each value.** Change nothing until all are recorded.

Param	What	Value
P-01	Max frequency (default 50)	
P-03	**Accel time, s** (default 5.0)	
P-04	Decel time, s (default 5.0)	
P-06	Energy optimisation (should be 0)	
P-07	Motor voltage = nameplate	
P-08	Motor current = nameplate	
P-09	Motor frequency = nameplate	
P-11	Torque boost % (default 3.0)	
P-12	Control mode (0 = terminals)	
P-38	Parameter lock	

- [ ] **Signs of a factory reset:** P-03 = 5.0, P-04 = 5.0, P-01 = 50.0 and P-08 not matching the nameplate. A kick lasts only 0.2–0.9 s, so with a 5 s ramp the motor never gets past about 10 Hz. **This alone explains a weak kick.**
- [ ] If P-03 is long: set it to **0.3 s** and watch a kick. If it then trips `O-I`, raise it in steps (0.5, 0.8 s).
- [ ] If P-06 = 1, set it to 0. It lowers motor voltage and torque.
- [ ] Speed setpoint: is there a speed pot on the drive or panel? Check it hasn't been turned down.
- [ ] Extra readouts: set **P-14 = 101**, then `P00-03` speed setpoint, `P00-10` run hours, `P00-17` current before the last trip.

D · Brake (power on)

- [ ] Stand by the wall when K1 (BRAKE RELEASE) clicks. A healthy brake gives a sharp **clack** on release. No clack means it isn't releasing.
- [ ] Wall wart output, unloaded: _____ V (expect about 24 V)
- [ ] Voltage at the brake leads with K1 on: _____ V (should stay above about 21.5 V)

Reading	Cause
0 V at brake, wall wart OK	K1 contact or wiring/connector
Voltage sags under load	Wall wart weak → replace
24 V present, no clack	Brake coil open, or air gap worn → F

Quick proof: power the brake straight from the wall wart, bypassing the relay, then run kicks. If the arm now swings properly, the fault is in the brake supply path.

E · Motor wiring (mains OFF)

Mains off, wait 10 min, then confirm 0 V at L/N and U/V/W.

- [ ] VFD terminals U, V, W: tight, no discolouration or burn marks
- [ ] Resistance at the VFD output terminals (motor still connected):
  U–V _____ V–W _____ W–U _____ Ω
  All three should be **equal within about 5 %**. One high or open = broken phase (cable, connector, winding).
- [ ] Each of U, V, W to PE/earth: **OL** (no connection)
- [ ] Brake cable continuity, end to end
- [ ] Run (K2) wiring at the VFD: terminals 1 and 2 tight

F · After opening the wall

Mains off for anything you touch.

- [ ] **Slip test:** draw a pen line across the gearbox output shaft and the arm hub. Run some kicks, then look again. If the line has split, the arm is slipping: tighten the set screw, key or clamp.
- [ ] Coupling between the motor and gearbox: tight, nothing loose
- [ ] Motor terminal box: all connections tight, no burn marks, star/delta bridges as on the nameplate
- [ ] Brake leads intact. With 24 V applied directly the brake should click and free the motor shaft or fan.
- [ ] Brake drags even with 24 V on → air gap or wear, the brake needs adjusting or replacing
- [ ] Gearbox: oil leak, grinding, play at the output shaft

Photo of the motor nameplate (brake code, coil voltage and wattage, rated current): ☐

G · Once it runs again

- [ ] Photograph **every** parameter P-01 to P-40. There is no record of the original settings yet.
- [ ] Note the kick values: peak Hz _____ peak A _____
- [ ] Note what you fixed: _____
- [ ] Close up the wall only after a long test run (several kick and pause cycles)

VFD: VersiDrive i E2 (peter electronic). Motor: SEVA-tec E series with 24 V spring brake. Old controller: K1 = brake release, K2 = VFD run (terminals 1–2).